

A Multimodal Vision-Language Reasoning & Stochastic Optimization Framework in Virtual Reality Neurosurgery Research

Cartik Sharma, Department of Surgical Engineering, Sunnybrook Health Sciences Centre

Abstract

This publication examines the implementation of 'shsc_vr_genai', a unified virtual-reality neurosurgical and orthopaedic training platform engineered for Sunnybrook Health Sciences. By mapping subcortical stereotactic target paths as stochastic Wiener drift processes and pedicle cortex drill profiles as high-density torque fields, we resolve trainee performance boundaries. Furthermore, we implement a Vision-Language-Action Multi-modal reasoner (VLM) which projects convolutional attention fields on active risk boundaries. Sequential resident competency standards are formally verified via sequential Wald SPRT stopping matrices to deliver automated, provably optimal certification bounds in surgical simulator environments.

1. Subcortical Cannulation & Wiener Drift Optimization

We model the surgeon's hand tool alignment as an attractor drift process. The micro-tremor coordinates solve a stochastic differential path attracted to targeted subthalamic coordinates:

$$dx_t = -\gamma(x_t - x_{\text{target}})dt + \sigma dW_t$$

where γ represents targeting stabilizer stiffness (a function of repetition reps) and dW_t is classical Brownian motion. Corresponding brain shear stress is calculated over parenchymal manifolds:

$$\text{Shear (kPa)} = 0.35 \cdot (\text{Deviation} \cdot 15.0)^{1.4} + 0.1 \cdot \text{drift}$$

2. Cortical-Cancellous Drilling Physics

Orthopaedic intramedullary force feedback models resistive drill torque over dense outer bone cortical layers, cancellous trabeculae, or osteoporotic cavities. Seating torque resistance (T_d) is resolved continuously:

$$T_{\text{seat}}(d) = 0.005 \cdot F_{\text{axial}} \cdot \rho(d) \cdot \frac{1000}{\sqrt{\text{RPM}}} + 0.002 \cdot \theta_{\text{tilt}} \cdot \left(\frac{d}{10.0}\right)$$

3. Spatial Convolutional Attention & Wald SPRT Inversion

VLM spatial representation parsing is facilitated by generating multi-headed circular attention vectors:

$$U_{\text{att}}(x_i, y_i) = \alpha \cdot \exp\left(-\frac{(x_i - x_c)^2 + (y_i - y_c)^2}{\sigma_{\text{att}}^2}\right)$$

Surgical resident evaluation is mathematically verified using Wald Sequential Probability Ratio boundaries to determine if the candidate qualifies for board certification under tight confidence limits ($\alpha = \beta = 0.05$):

$$\Lambda_k = \sum_{i=1}^k \ln\left(\frac{p(\text{skill}_i | H_1)}{p(\text{skill}_i | H_0)}\right) \geq 2.944 \Rightarrow \tau^* = \inf\{k : \Lambda_k \geq A\}$$

Sunnybrook Virtual Reality Surgery Simulator Analytics Matrix

